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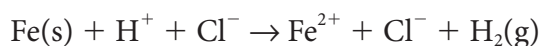
Class: _____

ACTIVITY SHEET

5.2 Analysing chemical reactions

Part A

Powdered iron reacts with hydrochloric acid to give soluble iron(II) chloride and hydrogen gas. The unbalanced equation for the reaction is shown below.



The iron reacts completely with 4 mol of hydrochloric acid.

Instruction

Complete the following tasks. (Refer to Appendix 2 in your textbook for data.)

- 1 State the relative atomic mass of iron.

$$A_r(\text{Fe}) =$$

- 2 Determine the relative atomic mass of hydrogen ions and chloride ions.

$$A_r(\text{H}^+) =$$

$$A_r(\text{Cl}^-) =$$

- 3 Calculate the relative molecular mass of hydrogen gas.

$$M_r(\text{H}_2) =$$

- 4 Rewrite the above equation so that it is balanced.

- 5 Determine the reaction's molar ratio from the coefficients and write the ratio under the equation.

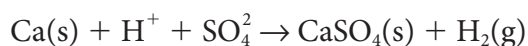
- 6 Use the molar ratio to determine the number of moles of iron needed to completely react with 4 mol of hydrochloric acid.

- 7 Determine the mass (m) of the iron in that number of moles.

- 8 Use the molar ratio to determine how many moles of hydrogen gas are released.
- 9 Determine the mass (m) of the hydrogen gas in that number of moles.

Part B

Powdered calcium metal reacts with sulfuric acid to produce insoluble calcium sulfate and hydrogen gas. The unbalanced equation for the reaction is shown below.



The calcium reacts completely with 0.5 mol of sulfuric acid.

Instruction

Complete the following tasks. (Refer to Appendix 2 in your textbook for data.)

- 1 State the relative atomic mass of calcium metal and the hydrogen ion.
 $A_r(\text{Ca}) =$
 $A_r(\text{H}^+) =$
- 2 Determine the relative formula mass of the sulfate ion and the calcium sulfate precipitate.
 $M_r(\text{SO}_4^{2-}) =$
 $M_r(\text{CaSO}_4) =$
- 3 Calculate the relative molecular mass of hydrogen gas.
 $M_r(\text{H}_2) =$
- 4 Rewrite the above equation so that it is balanced.
- 5 Determine the reaction's molar ratio from the coefficients and write the ratio under the equation.
- 6 Use the molar ratio of the reaction to determine the number of moles of calcium needed to completely react with 0.5 mol of sulfuric acid.
- 7 Determine the mass (m) of the calcium in that number of moles.

- 8 Use the molar ratio to determine how many moles of hydrogen gas are released.
- 9 Determine the mass (m) of the hydrogen gas in that number of moles.